Study of Voltage Control Current Source in Electrical Impedance Tomography System

Li Zhangyong, Xu Zhui, Ren Chaoshi, Wang Wei, Zhao Dechun, Zhang Huiquan
Biomedical Engineering Research Center
Chongqing University of Posts and Telecommunications
Chongqing, China
E-mail: zhuizhui2008@yahoo.com.cn

***Abstract*—A steady and efficient Voltage Control Current Source (VCCS) with broad bandwidth plays a very important role in the quality of final images for Electrical Impedance Tomography (EIT) system. Therefore, two models of current source are proposed in the paper: Advanced Howland VCCS and Double Operational Amplifiers VCCS. Both of the two models are simulated through a software named Multisim, and the results of simulation show that both models can keep an output impedance higher than 1MΩ when the frequency changes from 10 kHz to 1 MHz and the load varies from 100Ω to 1000Ω.The results also show that both models meet the requirements of VCCS in EIT system, they are hoped to provide usefull approaches for the design of practical VCCS of EIT system.**

Keywords-; Electrical Impedance Tomography; Howland circuit;double oprational amplifier current source circuit

I. INTODUCTION

EIT technology is based on the various impedance in different organizations of the human body, it can make full use of the physiological and pathological information which carried by human body impedance, a small and safety current is injected into human body by stimulating electrode, then voltage can be measured by measuring electrode, at last the voltage information is rebuilt into images that describe the electrical characteristics inside human body through a certain calculating algorithm[1]. EIT has been widely concerned because of its three advantages: no harmful to body, functional imaging and fitting for long-term medical care and it is also one of major topics of modern bio-informatics and biomedical engineering.

The research of EIT began in the 70s of last century; the EIT technology has made considerable progress in just 30s. Some groups and organizations have worked on the research of the clinical applications in some fields like: detection of breast cancer, brain and abdominal bleeding custody, measurement of lung function and so on [2][3][4]. It is time for the transition from theoretical research to clinical application.

EIT system includes software and hardware components: software component focuses on algorithm and imaging; hardware consists of Digital oscillator circuit, Voltage Control Current Source (VCCS), Control and data acquisition system, Voltage measurement circuit and Electrode array. VCCS plays a great important role in the process of imaging and the quality of final images, so the design of VCCS in EIT system is especially important [5]. An ideal VCCS in EIT system requires high precision and stability which means approximately infinite output impedance. In this paper, two models of current source are proposed: Advantaged Howland (AH*)* circuit and Double Operational Amplifier (DOA) current source circuit, both of the two models are analyzed and simulated, and the results of analysis and simulation are processed and compared to find a better method.

II. MATERIAL AND METHODS

A. Howland Circuit

The V-I conversion can be achieved when Current Series Negative Feedback (CSNF) were added into amplifier circuit, Howland circuit includes both CSNF and Voltage Positive Feedback (VPF), So it is a more practical current source circuit as is shown in Fig.1. When load resistance R_L decreases, on the one hand, i_O will increase because of the inner resistance; on the other hand, u_P will decrease, then u_O will decrease and i_O decrease too; When the value of resistances meet (1), the increased value causes by the reduction of R_L approximately equal to the decreased value causes by VPF, then V-I conversion is finished and a steady current computed by (2) can be given.

$$\frac{R_2}{R_1} = \frac{R_3}{R_4}, \tag{1}$$

$$i_O \approx \frac{-u_I}{R_4}. \tag{2}$$

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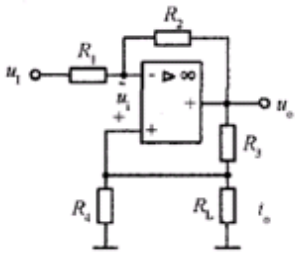


Fig.1 Howland circuit

B. VCCS Based on AH Circuit

VCCS based on AH circuit is shown in Fig.2. A feedback resistor R_4 is added to Howland circuit which helps to reduce the influence of current source causes by the variation of load resistance, so the output impedance is increased furthermore [6].

As is shown in Fig.2, when the value of resistances meet (3):

$$\frac{R_2}{R_1} = \frac{R_4 + R_5}{R_3}. \tag{3}$$

Then the output current i_L is computed by (4):

$$i_L \approx \frac{V_1}{R_3 // R_4}. \tag{4}$$

Equation (4) shows that the value of i_L is related to signal voltage V_1, periphery resistances R_3 and R_4 but is not related to load R_L

C. VCCS Based on DOA Current Source Circuit

DOA current source circuit is shown in Fig.3. A negative feedback is added to the inverting input port of U1, the in-phase input port is connected with a voltage follower (U2) and R_4 which also compose a sum operational circuit.

Then the output current can be calculated by (7):

$$u_{P1} = \frac{R_4}{R_3 + R_4} V_1 + \frac{R_3}{R_3 + R_4} u_{P2} \tag{5}$$

$$u_{O1} = (1 + \frac{R_2}{R_1}) u_{P1}. \tag{6}$$

Let $R_1 = R_2 = R_3 = R_4$,

$$\text{Then: } i_L = \frac{V_1}{R_0}. \tag{7}$$

Equation (7) shows that the value of i_L is related to signal voltage V_1, periphery resistance R_0 but is not related to load R_L.

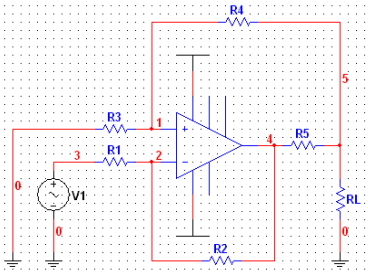


Fig.2 Advanced Howland circuit

D. VCCS In EIT System

The principle of EIT technology requires: the more steady the output current is, the closer the obtained impedance information approach to the truth; a wide frequency rage between 10 kHz and 1 MHz is required in order to obtain the information of imaginary part in impedance [7][8]; the output current must be smaller than 5mA for the consideration of the security of human body.

So far, several models of VCCS have been proposed: DOA VCCS, AH VCCS, Mirror Current (MC) VCCS and so on. The DOA model increases the output impedance of current source by adding a voltage follower with high input impedance to the feedback circuits; it is used more wildly but only applied to a bandwidth between 0 and 200 kHz which is lower than the requirement of EIT. The AH model needs only single operational amplifier and several resistor to obtain a high output impedance [9], so the selection of operational amplifier is of great importance. The MC model has a strict requirement on the symmetry of the current mirror. But thanks to the developments of modern electronic technology and integrated circuit manufacturing, some operational amplifiers integrate current mirror itself and the bandwidth became broader and broader.

So, current feedback operational amplifier AD844 is used in the design. The AD844 has several features like: wide bandwidth (60MHz), fast settling *(*100 ns to 0.1%*)*, slew rate (2000 V/s*)*; quickly response to signal, low distortion, low noise, low temperature offset, excellent performance on AC/DC; current mirror is integrated inside[10]. So the DOA VCCS with the application of AD844 owns both the features of DOA VCCS and advantages of current mirror, and the problem of insufficient bandwidth can be overcame.

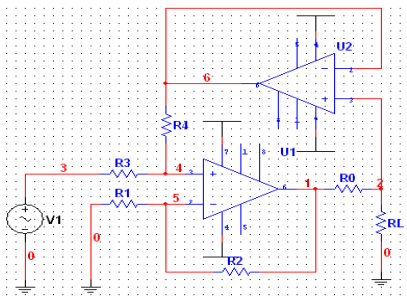


Fig.3 DOA current source circuit

III. RESULTS AND CONCLUSION

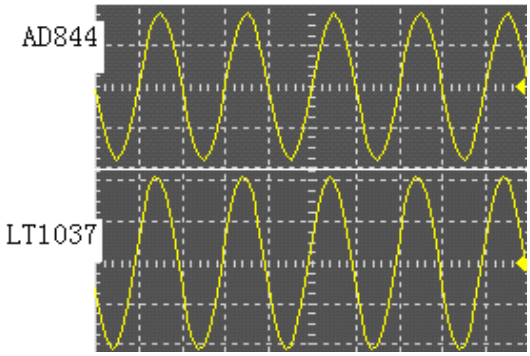
Both of the two models are simulated in Multisim, AD844 is used in the two models. The output waveform is observed and the output impedance is calculated after finishing setting all the parameters required.

Firstly, output waveform is observed when the output frequency varies from 10 kHz to1 MHz. Another chip LT1037 is chosen to do the same actions for comparison. The waveform is shown in Fig.4.

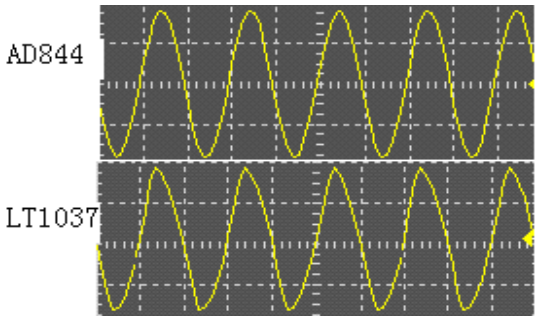
Secondly, the method of measuring the output impedance is presented in Fig.5. When the frequency changes from 10 kHz to 1 MHz and R_P changes from 100Ω to 1000Ω ,Voltages of R_1 were recorded when the switch is open or closed (switch opened- V_1 , switch closed- V_2), then the output impedance Z_{out} can be calculated by (8) [11][12], and the results are listed in Table 1.

$$Z_{out} = \frac{1}{V_2/V_1 - 1}(R_P - R_1) . \quad (8)$$

As is shown in Fig.4, both of AD844 and LT1037 provide excellent performances at 10 kHz when they are applied to AH circuit. When the frequency increases to 1 MHz, AD844 can still keeps steady output waveform but LT1037 can't, obvious distortion can be observed.

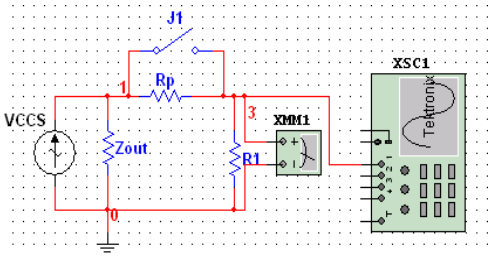


(a) The output current of AD844 and LT1037 applied to AH circuit at 10 kHz

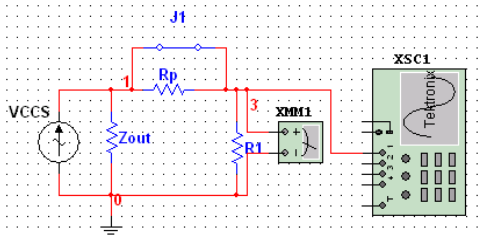


(b) The output current of AD844 and LT1037 applied to AH circuit at 1 MHz

Fig.4.The output current waveform at different frequencies.



(a) switch opened



(b) switch closed

Fig.5.The principle of measuring output impedance and waveform

TABLE 1.

THE OUTPUT IMPEDANCE OF BOTH MODELS WHEN THE FREQUENCY VARIES FROM 10 kHz TO 1 MHz

Frequency（Hz）	Double Amp Zout（Ω）	Advanced Howland Zout（Ω）
1M	1.6M	1M
500k	5.8M	3.6M
200k	21M	11M
100k	30M	18M
10k	39M	22.5M

As is shown in TABLE 1, both of the two models can keep high output impedance at the frequency ranging from 10 kHz to 1 MHz with the load varies from 100Ω to 1000Ω, and when the frequency is 1 MHz and the load is 1000Ω, the output impedance of AH model is 1MΩ and DOA model is 1.6 MΩ. It has been proved that the output impedance decreases dramatically when the frequency and the load increases, so only impedance in 1000Ω is presented in TABLE 1.

IV. DISCUSSION

In medical EIT system, the load is normally less than 100Ω and the frequency changes from 10 kHz to 1 MHz [13], and the output impedance must be higher than 100kΩ [14].Both of the two models perform well at high frequency when AD844 is used, and DOA model is better than AH model according to the results of simulation and analysis.

A steady and efficient VCCS requires not only high output impedance but also a stable phase, but only output impedance was discussed in this paper, how to keeps a stable phase is the task of further study.

The design of practical VCCS for EIT system is an important factor for the quality of final results. It is a problem that must be solved for the research of EIT. Therefore, the models mentioned above are hoped to provide references for the design of VCCS in EIT system.

ACKNOWLEDGMENT

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